ELEVEN, TWELVE, AND THIRTEEN LONELY RUNNERS

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Abstract. Wills conjectured that, for any integers $u_1, \ldots, u_k$, there is a real number t such that, for all $i = 1, \ldots, k$,

$$\|tu_i\| \geq \frac{1}{k+1},$$

where $\|x\|$ is the distance from x to the closest integer. This statement is known as the Lonely Runner Conjecture. A computational method developed by Rosenfeld and the second author verified the conjecture for $k \leq 9$. We further refine this method with new sieving techniques and employ a polynomial technique to show that any $(u_1, \ldots, u_k) \equiv (1, 2, \ldots, k) \pmod{p}$ with $\gcd(u_1, \ldots, u_k) = 1$ satisfies the conjecture when $k+1$ and $p > k^2 + k$ are both odd primes. Ultimately, we provide a computer-assisted proof of the Lonely Runner Conjecture for $k \in \{10, 11, 12\}$.

1. Introduction

The Lonely Runner Conjecture arose in the study of Diophantine approximation [16], though it has been reinterpreted in the context of view obstruction and billiard trajectories [5], covering radii of zonotopes [8], and the chromatic number of a unit-distance graph [17]. It was conjectured independently by Wills [16] in 1967 and Cusick [5] in 1984. By writing $\|x\|$ for the distance from x to the closest integer, namely $\|x\| = \min(x - \lfloor x \rfloor, \lceil x \rceil - x)$, the conjecture can be stated as follows:

Conjecture 1.1. *Let $u_1, \ldots, u_{k+1}$ be distinct real numbers. For each $i \in \{1, \ldots, k+1\}$, there is a real number t such that*

$$\|tu_i - tu_j\| \geq \frac{1}{k+1} \quad \textit{for all } j \neq i.$$

Conjecture 1.1 has a popular interpretation that gives its name: if $k+1$ runners run with distinct, constant speeds around a unit-length track, each of them will be *lonely* at some time. Being lonely means being at least $\frac{1}{k+1}$ away from other runners. However, most advances work with one fewer runner and only positive integer speeds, possible due to [4] and [8] that showed the equivalence between Conjecture 1.1 and the following statement:

Conjecture 1.2. *All $(u_1, \ldots, u_k) \in \mathbb{Z}^k_{>0}$ have the* LR property*, meaning that there is a real number t such that*

$$\|tu_i\| \geq \frac{1}{k+1} \quad \textit{for all } i = 1, \ldots, k.$$

The time t where the above inequalities hold is called a witness *of the LR property.*

We denote this statement by LRC(k) or, as customary, the Lonely Runner Conjecture for $k+1$ runners.

Conjecture 1.2 is known to hold if we replace $\frac{1}{k+1}$ with $\frac{1}{2k} + \frac{1}{k^{5/3+o(1)}}$ [2], or, more closely related to our approach, for $k \leq 9$ (at most 10 runners). The two smallest values of k are trivial. Betke and Wills [3] first proved the case $k = 3$, Cusick and Pomerance [5] proved $k = 4$, and Bohman, Holzman, and Kleitman [4] proved $k = 5$. Barajas and Serra [1] settled the case $k = 6$ in 2008.

This had remained the largest verified case prior to September 2025, when Rosenfeld [11] introduced a new framework to handle $k = 7$ (and later $k = 8$ [12]). He demonstrated that, with computer assistance, the conjecture can be verified in a number of small ansatze. In

Date: April 16, 2026.
2020 *Mathematics Subject Classification.* 11K60.

doing so, he relied on the finite-checking result of Malikiosis, Santos, and Schymura [9], which follows from a similar result of Giri and Kravitz [6]. The second author [14] then, in November 2025, presented a sieving scheme that substantially reduces the work for verification, thereby concluding $k \in \{8, 9\}$.

The main objective of this paper is to prove $LRC(k)$ for $k \in \{10, 11, 12\}$.

Theorem 1.3. *$LRC(k)$ holds for $k \leq 12$.*

In order to carry out the proof, we improve upon the framework of [11] and [14] both mathematically and computationally. We refine the sieving scheme with extra intermediate sieves and exploit a modulo p symmetry during implementation to reduce the verification necessary. These were enough to prove $k = 11$. Furthermore, we employ the polynomial method to establish the following proposition that finishes the proof for $k \in \{10, 12\}$.

Proposition 1.4. *Let $(u_1, \ldots, u_k) \in \mathbb{Z}^k_{>0}$ and $p > k^2 + k$ where both p and $k+1 \geq 3$ are primes. If $\gcd(u_1, \cdots, u_k) = 1$ and*

$$u_i \equiv i \pmod{p} \quad \textit{for all } i = 1, \ldots, k,$$

then $(u_1, \ldots, u_k)$ has the LR property.

Our proof strategy is outlined in the next section. Section 3 discusses the intermediate sieves technique and Section 4 is devoted to proving Proposition 1.4. We describe the verification algorithms and their implementation in Section 5, the results of which establish Theorem 1.3 in Section 6. Lastly, we discuss in Section 7 how this method may be further extended.

We adopt the following choices for notation. We write $aS = \{as : s \in S\}$ for $a \in \mathbb{R}$ and $S \subseteq \mathbb{R}$. The additive group modulo n is denoted $\mathbb{Z}_n$, the multiplicative group modulo n is $\mathbb{Z}_n^\times$, and

$$\mathbb{Z}_{p,l} := \mathbb{Z}_{pl} \setminus p\mathbb{Z}_l.$$

Let $\pi_{m\to n} : \mathbb{Z}_m \to \mathbb{Z}_n$ denote the natural projection map when $n \mid m$ and $\pi_n : \mathbb{Z} \to \mathbb{Z}_n$ denote the projection modulo n of integers. The fractional part of $x \in \mathbb{R}$ is denoted by $\{x\}$.

We use boldface letters for vectors, subscripts to indicate coordinates of a vector, and $\widehat{\cdot}$ to indicate the omission of a certain coordinate in a vector. For example, u_i is the ith coordinate of the vector $\mathbf{u}$. We refer to a vector $\mathbf{u}$ or $\mathbf{v}$ as a *speed tuple* to match the conjecture's settings. The application of a scalar function to vectors is coordinate-wise.

2. Proof strategy

Rosenfeld's [11] key insight is that if $\mathbf{u} \equiv \mathbf{v} \pmod{lp}$, then $\|tv_i\| = \|tu_i\|$ for every i whenever t is a rational with denominator lp. In particular, if we know that $\mathbf{v} \in \mathbb{Z}^k_{lp}$ has a witness time in the form $t \in \frac{1}{lp}\mathbb{Z}$, then every $\mathbf{u} \in \pi^{-1}_{lp}(\mathbf{v})$ has the same witness time and thus the LR property. If every $\mathbf{v} \in \mathbb{Z}^k_{p,l}$ has a witness time in the form $t \in \frac{1}{lp}\mathbb{Z}$, then one of the speeds in a counterexample to $LRC(k)$ must be divisible by p. The set of speed tuples $\mathbb{Z}^k_{p,l}$ together with the set of times $\frac{1}{lp}\mathbb{Z}$ is what we informally call an *ansatz*. The following definition and lemma summarize what must be verified.

Definition 2.1 (Definition 3.1[1], [14])**.** Let p be prime, $k \geq 2$ and l be positive integers. A speed tuple $\mathbf{v} \in \mathbb{Z}^k_{p,l}$ is (k, p, l)-*proper* if one of the following conditions holds:

- there exists i such that $\gcd(l, v_1, \ldots, \widehat{v_i}, \ldots, v_k) > 1$.
- there exists $t \in \frac{1}{lp}\mathbb{Z}$ such that $\|tv_i\| \geq \frac{1}{k+1}$ for all i.

If $\mathbf{v}$ is not proper, it is *improper*. Moreover, let $I(k, p, l)$ denote the set of all improper speed tuples.

Lemma 2.2 (Lemma 3.3, [14])**.** *For $k \geq 3$, if $LRC(k-1)$ is true and $I(k, p, l)$ is empty for some l, then any counterexample $\mathbf{u}$ to $LRC(k)$ satisfies $p \mid u_1 \cdots u_k$.*

[1]We change lp in the gcd condition of [14] to l, but this changes nothing when p is prime.

Lemma 2.2 translates proving Conjecture 1.2 to the computer verification task of showing $I(k,p,l_p) = \emptyset$ for some l_p, for many primes p. Since the choice of l is irrelevant to the final conclusion, we introduce a new terminology to simplify our discussion and implementation.

Definition 2.3. Let $k \geq 2$ be an integer and p be a prime. A speed tuple $\mathbf{v} \in \mathbb{Z}^k_{p,1}$ is said to be *eventually* (k,p)*-proper* if there is l such that

$$\mathbf{v} \notin \pi_{lp\to p} I(k,p,l).$$

Let $J(k,p)$ denote the set of all $\mathbf{v} \in \mathbb{Z}^k_{p,1}$ that are not eventually (k,p)-proper.

Eventual properness might appear to be a local property of each speed tuple $\mathbf{v}$, but having this local properties throughout every $\mathbf{v}$ implies the global conclusion that $I(k,p,l) = \emptyset$ for some l.

Lemma 2.4. *For fixed k and p, $J(k,p) = \emptyset$ if and only if $I(k,p,l) = \emptyset$ for some l.*

Proof. The backward direction is immediate from the definition. To prove the forward direction, note that $J(k,p) = \emptyset$ implies that, for each $\mathbf{v} \in \mathbb{Z}^k_{p,1}$, there is $l_{\mathbf{v}}$ for which $\mathbf{v} \notin \pi_{l_{\mathbf{v}}p\to p} I(k,p,l_{\mathbf{v}})$. Then pick

$$l = \operatorname{lcm}\{l_{\mathbf{v}} : \mathbf{v} \in \mathbb{Z}^k_{p,1}\}.$$

Fix $\mathbf{u} \in \mathbb{Z}^k_{p,l}$. Let $\mathbf{v} = \pi_{lp\to p}\mathbf{u}$ and $\mathbf{v}' = \pi_{lp\to l_{\mathbf{v}}p}\mathbf{u}$, so we can write

$$\mathbf{u} = (a_1 l_{\mathbf{v}} p + v'_1, \ldots, a_k l_{\mathbf{v}} p + v'_k)$$

for some integers $a_1, \ldots, a_k$.

Notice that $\pi_{l_{\mathbf{v}}p\to p}\mathbf{v}' = \mathbf{v}$. By definition of $l_{\mathbf{v}}$, $\mathbf{v}'$ is $(k,p,l_{\mathbf{v}})$-proper. If $\mathbf{v}'$ satisfies the gcd condition, then there is i such that there is $d := \gcd(l_{\mathbf{v}}, v'_1, \ldots, \widehat{v'_i}, \ldots, v'_k) > 1$. It follows that $d \mid l$ and $d \mid u_j$ for all $j \neq i$, hence $\mathbf{u}$ also satisfies the gcd condition.

Otherwise, there is $t \in \frac{1}{l_{\mathbf{v}}p}\mathbb{Z}$ such that $\|tv'_i\| \geq \frac{1}{k+1}$ for all i. Then observe that $\frac{1}{l_{\mathbf{v}}p}\mathbb{Z} \subseteq \frac{1}{lp}\mathbb{Z}$, so $t \in \frac{1}{lp}\mathbb{Z}$, and $\|tu_i\| = \|tv_i\| \geq \frac{1}{k+1}$ for all i.

In both cases, we have shown $\mathbf{u}$ to be (k,p,l)-proper. Therefore, $I(k,p,l) = \emptyset$. □

As a consequence, the emptiness of $J(k,p)$ and $I(k,p,l)$ at some l may be discussed interchangeably. We may also restate Lemma 2.2 in a more usable form for the proof of our main Theorem 1.3.

Corollary 2.5. *Let $k \geq 3$ and assume that $LRC(k-1)$ is true. If $J(k,p)$ is empty, then any counterexample $\mathbf{u}$ to $LRC(k)$ has $p \mid u_1 \cdots u_k$.*

Finally, we require an upper bound on the runners' speeds beyond which the LR property is guaranteed. The best such bound at present was obtained by Malikiosis, Santos, and Schymura [9]:

Lemma 2.6 (Corollary 3, [11])**.** *If $LRC(k-1)$ holds, then any counterexample $\mathbf{u}$ to $LRC(k)$ must have*

$$u_1 \cdots u_k < B_k \quad \textit{where} \quad B_k := \left(\frac{\binom{k+1}{2}^{k-1}}{k} \right)^k.$$

Therefore, if we satisfy the precondition of Corollary 2.5 for a large number of primes, we can obtain a lower bound of the speeds in any counterexamples. Together with stated upper bound, we can conclude the absence of a counterexample.

For convenience, we consolidate the verification task into the following proposition, which will be used to conclude the main theorem.

Proposition 2.7. *Let $k \geq 3$ and P be a set of primes such that $J(k,p) = \emptyset$ for all $p \in P$. If $LRC(k-1)$ holds and*

$$\prod_{p \in P} p \geq B_k.$$

Then, $LRC(k)$ is true.

Proof. Assume that there exists a counterexample $\mathbf{u}$ to $LRC(k)$. By Corollary 2.5, $\prod_{p\in P} p$ divides $u_1 \cdots u_k$ and so

$$u_1 \cdots u_k \geq \prod_{p\in P} p \geq B_k > u_1 \cdots u_k,$$

where the last inequality comes from Lemma 2.6. This is a contradiction. □

3. Sieving techniques

In order to carry out our proof strategy, we need to verify that $J(k,p) = \emptyset$. The first approach is to check all tuples in $\mathbb{Z}^k_{pl}$ to verify that $I(k,p,l) = \emptyset$, but this entails checking approximately $(pl)^k$ tuples. Moreover, for reasons to be given in Remark 3.2, we need $l \geq k+1$, so this approach becomes very laborious even for $k = 8$ [12].

Before we motivate them, let us at once formalize and validate the two techniques that allows for efficient verification in the following proposition.

Proposition 3.1. *Let $S \subseteq \mathbb{Z}^k_{p,l}$ be such that $J(k,p) \subseteq \pi_p S$. We also have $J(k,p) \subseteq \pi_p S'$ if S' is constructed in one of the following ways:*

- *(lifting with c) $S' = \{\mathbf{v} \in \pi^{-1}_{clp\to lp} S : \mathbf{v}$ is (k,p,cl)-improper$\}$*
- *(projecting) $S' = \pi_p S$.*

Proof. The second case is obvious. In the first case, note that

$$\begin{aligned}\pi_p S' &= \pi_p\left(\pi^{-1}_{clp\to lp} S \cap I(k,p,cl)\right)\\ &= \pi_p \pi^{-1}_{clp\to lp} S \cap \pi_p I(k,p,cl)\\ &= \pi_p S \cap \pi_p I(k,p,cl).\end{aligned}$$

As $J(k,p)$ is a subset of $\pi_p S$ by assumption and a subset of $\pi_p I(k,p,cl)$ by definition, it is also a subset of $\pi_p S'$.[2] □

The two actions are lifting, which we generalize from [14], and backward projection, a new technique. Proposition 3.1 ensures that both actions preserve the inclusion of $J(k,p)$ (under projection), which means that we cannot lose a non-eventually proper speed tuple in the process. Now, we will motivate each of them by their applications.

3.1. Lifting.

Fix k,p and assume that the computational cost of checking each speed tuple is equal to 1.

Lifting is the main innovation of [14]. The process is described in the construction of S' in Proposition 3.1: suppose we have a tuple in $\mathbb{Z}^k_{p,l}$, we can *lift* it to the preimage in $\mathbb{Z}^k_{p,cl}$, and then discard all proper speed tuples.

Remark 3.2. The speed tuple $(1,2,\ldots,k)$ is lonely precisely when $t = \frac{s}{k+1}, \gcd(s,k+1) = 1$. Therefore, if $p > k+1$ and $k+1 \nmid l$, then there is no such time in $\frac{1}{pl}\mathbb{Z}$, so $(1,2,\ldots,k) \in I(k,p,l)$. In other words, if $p > k+1$, then $I(k,p,l) = \emptyset$ only if $k+1 \mid l$. As a consequence, if $k+1$ is prime and we start with tuples in $\mathbb{Z}^k_p$ we need to lift with $k+1$ (or a multiple of $k+1$, but this would be unoptimal) at least once because we can only eliminate $(1,2,\ldots,k)$ only once we get to $\mathbb{Z}^k_{p(k+1)}$

Even though it is impossible that $I(k,p,1) = \emptyset$, in view of applying Proposition 3.1, we may still compute $I(k,p,1)$ first. This is because $I(k,p,1)$ is a non-trivial *upper bound*, a superset under projection, of $J(k,p)$ that is the easiest to compute as it requires checking only approximately p^k speed tuples. After obtaining that, we can lift with a divisor of $k+1$ to obtain a better upper bound.

[2] This proof looks deceptively simple, and perhaps curiously, does not utilize the lifting lemma (Lemma 3.5 [14]) anywhere. This is because the real work (and the very same work as the lifting lemma) has been done in Lemma 2.4 to show that $J(k,p) = \emptyset$ is indeed the same as $I(k,p,l) = \emptyset$ for some l. Without this fact, there is no divisibility result in Corollary 2.5 hence determining $J(k,p)$ is pointless.

Observe that lifting a set S of tuples with a multiplier c requires checking

$$\left|\pi_{clp\to lp}^{-1}S\right| = c^k \cdot |S|$$

speed tuples. As S is empirically found to be smaller as we lift[3], we should take c to be small at first. Furthermore, as k gets larger, the value of c has greater effect on the computational cost, and the $c = 2$ lift becomes comparatively cheaper.

The key improvement from [14] is noting that, in fact, we can take this idea to the extreme. Since the $c = 2$ lift is computationally cheaper to compute, we can apply a $c = 2$ lift first (and successively) whether or not 2 is a prime factor of k. This is done to trim down the number of speed tuples in S when we apply a more computationally expensive, but necessary lift. Even better, doing this will never increase the number of speed tuples at the end because we can always project backward, as we will explain next.

3.2. **Backward projection.** Backward projection is just the application of π_p to pull back the speed tuples to $\mathbb{Z}_p^k$. Just as lifting can reduce the size of the search space, projection achieves a similar effect. This is due to the fact that for any set S,

$$|\pi_p S| \leq |S|\,.$$

The only time projecting becomes counterproductive is if we lift afterwards with the same constant c that was previously used, because then we are intersecting with the same improper set and hence no speed tuples can be further eliminated.

Therefore, in general, we can lift with $c = 2$ successively, then project back, then lift with $c = 3$ successively and project back, and so on. This is the strategy that we use in our verification algorithms in Subsection 5.2.

4. Eventual properness of $(1, 2, \ldots, k)$

According to Remark 3.2, when $k+1$ is prime, we have to lift with the multiple $c = k+1$. This lift is extremely computationally expensive: for $k = 12$, the $c = k + 1$ lift is $(13/2)^{12} \approx 5 \cdot 10^9$ times more computationally expensive than the $c = 2$ lift. If we are to verify the conjecture for $k \in \{10, 12\}$ in a reasonable time, we need to avoid doing this lift computationally. For many primes p, after lifting $I(10, p, 1)$ with $c = 2, 3$ several times and projecting back, the only tuples that cannot be shown eventually proper are equivalents[4] of $(1, 2, \ldots, 10)$. The analogous observation holds when $k = 12$.

Therefore, the aim of this section is to determine the eventual properness of $(1, 2, \ldots, k)$. We emphasize that our method still requires *lifting* with $c = k + 1$. In fact, we merely are finding a solution for speed tuples in $\pi_{(k+1)p\to p}^{-1}(1, 2, \ldots, k)$ in the $(k, p, k + 1)$-ansatz, but we do so analytically instead of computationally.

First, we prove the following Proposition 4.1 which is the main ingredient used in this section. This proposition is combinatorially interesting in its own right. For convenience, we write

$$N_k := \left\{\mathbf{v} \neq \mathbf{0} \text{ and } \mathbf{v} \text{ has at least one zero coordinate } : \mathbf{v} \in \mathbb{Z}_{k+1}^k\right\}.$$

Proposition 4.1. *Let $k + 1$ be an odd prime. For all vectors $\mathbf{v} \in N_k$, there is s and r in $\mathbb{Z}_{k+1}^\times$ such that*

$$s\mathbf{v} + r(1, 2, \ldots, k) \in \{1, \ldots, k - 1\}^k.$$

Proof. In this proof, we work in the field $\mathbb{Z}_{k+1}$. Define

$$V_m = \{v_i + mi : i \in \{1, \ldots, k\}\}$$

for $m \in \mathbb{Z}_{k+1}^\times$. We say $m \in \mathbb{Z}_{k+1}^\times$ is *good* if $0 \notin V_m$ and *bad* otherwise. Let G be the set of good m's.

[3]Even though lifting increases the number of speed tuples to be checked, it seems that the increased precision of the allowed time in the lifted ansatz has a greater effect [14].

[4]Speed tuples that are eventually proper if and only if $(1, 2, \ldots, 10)$ is, to be made explicit in Subsection 5.1.

Now, as $\mathbf{v}$ contains at least one zero coordinate, it contains at most $k-1$ non-zero coordinates. For each non-zero coordinate v_i, there is exactly one choice of $m \in \mathbb{Z}_{k+1}^{\times}$ such that $v_i + mi = 0$, whereas for each zero coordinate, there cannot be such m. Thus, there are at most $k-1$ bad units. This implies that $|G| \geq 1$.

We continue the proof by contraposition. Assume that for all r and s,

$$s\mathbf{v} + r(1, 2, \ldots, k) \notin \{1, \ldots, k-1\}^k,$$

then we will show that $\mathbf{v} = 0$.

Let m be good and $s \in \mathbb{Z}_{k+1}^{\times}$. Then $r := sm \in \mathbb{Z}_{k+1}^{\times}$ and so, by assumption,

$$s\mathbf{v} + r(1, 2, \ldots, k) \notin \{1, \ldots, k-1\}^k.$$

Thus, there is an index i such that $s(v_i + mi) \in \{0, -1\}$. As m is good and $s \neq 0$, it is impossible for the product to be 0, and so it must be -1. Hence,

$$v_i + mi = -s^{-1}.$$

As s was chosen arbitrarily, $V_m = \mathbb{Z}_{k+1}^{\times}$.

From $V_m = \mathbb{Z}_{k+1}^{\times}$ when m is good, we have that

$$k! = \prod_{i=1}^{k}(v_i + mi) = \prod_{i=1}^{k} i \cdot \prod_{i=1}^{k}(v_i i^{-1} + m) = k! \prod_{i=1}^{k}(v_i i^{-1} + m),$$

which gives

$$\prod_{i=1}^{k}(v_i i^{-1} + m) = 1.$$

On the other hand, when m is bad, then there is an index i such that $v_i + mi = 0$, hence

$$\prod_{i=1}^{k}(v_i i^{-1} + m) = \prod_{i=1}^{k}(v_i + mi) i^{-1} = 0.$$

Therefore, the polynomial

$$P(X) := \prod_{i=1}^{k}(v_i i^{-1} + X)$$

is an indicator function of G on $\mathbb{Z}_{k+1}^{\times}$. Note also that since $\mathbf{v}$ contains at least one zero coordinate, $P(0) = 0$.

As $k+1$ is prime, Fermat's Little Theorem gives another indicator function of G, which is

$$Q(X) := \sum_{m \in G}(1 - (X - m)^k),$$

where the sum is not empty as $|G| \geq 1$. It also follows that Q is a polynomial of degree k.

As $P(0) = 0 = Q(0)$, P and Q are polynomials of degree at most k that agree on $k+1$ points, hence $P = Q$. In particular, their leading coefficients must be equal, that is $1 \equiv -|G| \pmod{k+1}$. Since $|G| \leq k$ as integers, we must have that $|G| = k$. Therefore, there is no bad m. Specifically, for any $i \in \{1, \ldots, k\}$, $v_i + mi \neq 0$ for all $m \in \mathbb{Z}_p^{\times}$. However, this implies $v_i = 0$ for all i, that is, $\mathbf{v} = \mathbf{0}$. □

The strategy we will later use to prove Proposition 4.4 is that we want to decouple and limit dependency of the prime p. Instead of considering times of the form $t = \frac{a}{(k+1)p}$ we decompose them into the partial fractions $t = \frac{s}{k+1} + \frac{r}{p}$. The former term has no dependence of p. To capture the effect of p of the latter term, we define

$$\mathbf{r}_k(t) := (\lfloor (k+1)\{t\} \rfloor,\ \lfloor (k+1)\{2t\} \rfloor,\ \ldots,\ \lfloor (k+1)\{kt\} \rfloor)$$

for $t \in \mathbb{R}$. Intuitively, $\mathbf{r}_k(t)_i = x$ means that the runner with speed i is the arc $\left[\frac{x}{k+1}, \frac{x+1}{k+1}\right)$ of the unit circle. With this discretization, we state Lemma 4.2.

Lemma 4.2. *Let $k+1$ be an odd prime. For every $\mathbf{v} \in N_k$, there is $s \in \mathbb{Z}_{k+1}$ and $r \in \mathbb{Z}$ for which*

$$s\mathbf{v} + \mathbf{r}_k(\tfrac{r}{k+1}) \in \{1, \dots, k-1\}^k$$

when considered modulo $k+1$.

Proof. Let $\mathbf{v}$ be any speed tuple in $\mathbb{Z}_{k+1}^k \backslash \{\mathbf{0}\}$ that contains a zero coordinate. By Proposition 4.1, there exist $s, r \in \mathbb{Z}_{k+1}^\times \subseteq \mathbb{Z}_{k+1}$ such that

$$s\mathbf{v} + r(1, 2, \dots, k) \in \{1, \dots, k-1\}^k.$$

By definition,

$$\begin{aligned}\mathbf{r}_k\left(\tfrac{r}{k+1}\right) &= \left(\left\lfloor (k+1)\left\{\tfrac{r}{k+1}\right\}\right\rfloor, \left\lfloor (k+1)\left\{\tfrac{2r}{k+1}\right\}\right\rfloor, \dots, \left\lfloor (k+1)\left\{\tfrac{kr}{k+1}\right\}\right\rfloor\right) \\ &= (r, 2r, \dots, kr) \\ &= r(1, 2, \dots, k)\end{aligned}$$

modulo $k+1$. Therefore,

$$s\mathbf{v} + \mathbf{r}_k\left(\tfrac{r}{k+1}\right) \in \{1, \dots, k-1\}^k,$$

modulo $k+1$. □

A corollary of Lemma 4.2 is that $(1, 2, \dots, k)$ is eventually $(k, k+1)$-proper, but this is trivial. Indeed, for $p = k+1$, we only need that

$$s\mathbf{v} + r(1, 2, \dots, k) \in \{1, \dots, k\}^k.$$

to show eventual (k, p)-properness of $(1, 2, \dots, k)$. However, proving the stronger statement that

$$s\mathbf{v} + r(1, 2, \dots, k) \in \{1, \dots, k-1\}^k$$

in Proposition 4.1 allows us to extend the result to larger primes via the next lemma.

Lemma 4.3. *Let $k+1$ and $p > k(k+1)$ be odd primes. For every $\mathbf{v} \in N_k$, there is $s \in \mathbb{Z}_{k+1}$ and $r \in \mathbb{Z}$ for which*

$$s\mathbf{v} + \mathbf{r}_k(r/p) \in \{1, \dots, k-1\}^k$$

when considered modulo $k+1$.

Proof. From Lemma 4.2, we have that there is $s \in \mathbb{Z}_{k+1}$ and $r \in \mathbb{Z}$ for which

$$s\mathbf{v} + \mathbf{r}_k(r/(k+1)) \in \{1, \dots, k-1\}^k$$

Thus, it suffices to show that $\mathbf{r}_k(\frac{1}{k+1}\mathbb{Z}) \subseteq \mathbf{r}_k(\frac{1}{p}\mathbb{Z})$ for all $p > k(k+1)$.

Fix p to be a prime greater than $k(k+1)$.

Notice that $\mathbf{r}_k$ is a piecewise constant function. Moreover, x is a discontinuity of $\mathbf{r}_k$ if and only if it is a discontinuity of $R_{k+1}(ix)$ for some i. Those points of discontinuity are precisely points of the form $\frac{a}{i(k+1)}$ for integer a and $1 \le i \le k$.

Choose an integer n and denote $x := \frac{n}{k+1}$. Observe that x is a discontinuity of $\mathbf{r}_k$ and let y be the smallest number greater than x that is also a discontinuity of $\mathbf{r}_k$. We know that y can be written as $\frac{a}{i(k+1)}$ for some integer a and $1 \le i \le k$, so it follows that

$$y - x = \frac{a}{i(k+1)} - \frac{n}{k+1} = \frac{a - ni}{i(k+1)} \ge \frac{1}{i(k+1)} \ge \frac{1}{k(k+1)}.$$

As y was chosen to be the smallest such discontinuity, $\mathbf{r}_k$ is constant on the interval $[x, y)$. Since the interval's width is at least $\frac{1}{k(k+1)} > \frac{1}{p}$, there must be an integer m such that $\frac{m}{p} \in [x, y)$. For such value m, $\mathbf{r}_k(\frac{n}{k+1}) = \mathbf{r}_k(\frac{m}{p})$. As n is arbitrary, the desired inclusion follows. □

With this lemma, we finally obtain the headline result that assures eventual properness of $(1, 2, \dots, k)$ for $p > k(k+1)$.

Proposition 4.4 (Restatement[5] of Proposition 1.4)**.** *Suppose that $k+1$ and $p > k(k+1)$ are odd primes. Then, every speed tuple in $\pi^{-1}_{(k+1)p\to p}(1,2,\dots,k)$ is $(k,p,k+1)$-proper, so in particular $(1,2,\dots,k)$ is eventually (k,p)-proper.*

Proof. We will show that every $\mathbf{u} \in \pi^{-1}_{p(k+1)\to p}(1,2,\dots,k)$ is $(k,p,k+1)$-proper, and hence $(1,2,\dots,k)$ is eventually (k,p)-proper. Let $u_i = a_i p + i$ for some $a_i \in \mathbb{Z}_{k+1}$ for $i = 1,\dots,k$. By considering a time of the form $t = \frac{s}{k+1} + \frac{r}{p}$ for $s \in \mathbb{Z}$ and $r \in \mathbb{Z}$, note that the fractional part of tu_i is given by

$$\{tu_i\} = \left\{\left(\frac{s}{k+1} + \frac{r}{p}\right)(a_i p + i)\right\} = \left\{\frac{su_i'}{k+1} + \frac{ri}{p}\right\}$$

where $\mathbf{u}' := \pi_{p(k+1)\to k+1}\mathbf{u}$.

If $\mathbf{u}' = \mathbf{0}$, then it follows that $k+1 \mid \gcd(\mathbf{u})$, thus

$$k+1 \mid \gcd(k+1, \mathbf{u})$$

hence $\mathbf{u}$ is $(k,p,k+1)$-proper because it satisfies the gcd condition.

If $\mathbf{u}'$ contains no zero coordinate, then pick $r = 0$ and $s = 1$ and we have

$$\{tu_i\} = \left\{\frac{u_i'}{k+1}\right\}$$

Since $u_i \neq 0$, $\|tu_i\| \geq \frac{1}{k+1}$ for all i.

Otherwise, $\mathbf{u}' \in N_k$, so by Lemma 4.3 there is a choice of r and s such that

$$s\mathbf{u}' + \mathbf{r}_k(r/p) \in \{1,\dots,k-1\}^k$$

when considered modulo $k+1$. This means that for each coordinate i, there are A_i and B_i such that $\{\frac{sv_i}{k+1}\} = \frac{A_i}{k+1}$ and $\{\frac{ri}{p}\} \in \left[\frac{B_i}{k+1}, \frac{B_i+1}{k+1}\right)$ where $C_i := A_i + B_i$ is in $\{1,\dots,k-1\}$ when considered modulo $k+1$. Therefore,

$$\{tu_i\} = \left\{\frac{su_i}{k+1} + \frac{ri}{p}\right\} \in \left[\frac{C_i}{k+1}, \frac{C_i+1}{k+1}\right)$$

and in particular the set on the right-hand side does not intersect with $[0, \frac{1}{k+1}) \cup (\frac{k}{k+1}, 1)$. Since this holds for every i, $\mathbf{u}$ has the LR property.

As there exists a witness time $t = \frac{s}{k+1} + \frac{r}{p} \in \frac{1}{p(k+1)}\mathbb{Z}$, $\mathbf{u}$ is $(k,p,k+1)$-proper. □

5. Implementation

5.1. **Equivalence of speed tuples.** It is clear that some speed tuples in an ansatz are essentially the same, so we only need to verify properness for one of them. We say that two speed tuples $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{Z}^k_{p,1}$ are *equivalent* if $\mathbf{v}$ can be obtained from $\mathbf{u}$ by performing a finite number of the following actions:

- Permuting the coordinates of $\mathbf{u}$
- Flipping the sign of a coordinate u_i to $-u_i$
- Multiplying $\mathbf{u}$ by some $a \in \mathbb{Z}_p^\times$.

This defines a group[6] action, hence the equivalence classes, i.e., the orbits, are well-defined. The first two actions are used in both [11] and [14], but the third is novel. For completeness, we show that this equivalence indeed preserves eventual properness.

Proposition 5.1. *Let p be prime and $\mathbf{u}, \mathbf{v} \in \mathbb{Z}^k_{p,1}$ be equivalent. If $\mathbf{u}$ is eventually (k,p)-proper, then so is $\mathbf{v}$.*

[5]Proposition 4.4 implies Proposition 1.4: suppose that $\gcd(\mathbf{v}) = 1$ and $\pi_p(\mathbf{v}) = (1,2,\dots,k)$, then we have shown in Proposition 4.4 that $\pi_{(k+1)p}(\mathbf{v})$ is $(k,p,k+1)$-proper. If it is proper due to gcd condition, then as $\gcd(\mathbf{v}) = 1$, by the pre-jump technique (see e.g. Lemma 5, [11]), $\mathbf{v}$ has the LR property. Otherwise, there is a time $t \in \frac{1}{(k+1)p}\mathbb{Z}$ such that $\|tv_i\| \geq \frac{1}{k+1}$ for all i.

[6]For the interested reader, the group is isomorphic to the direct product of $\mathbb{Z}_p^\times$ with the k-dimensional hyperoctahedral group.

Proof. We only need to show that each of the aforementioned actions preserves eventual properness. For permuting it is trivial. Suppose $\mathbf{u}$ is eventually proper and l is the integer for which every element in $\pi^{-1}(\mathbf{u})$ is proper, where $\pi := \pi_{lp\to p}$. Now let $\mathbf{v}' \in \pi(\mathbf{v})$.

If $\mathbf{v} = (u_1, \dots, -u_i, \dots, u_k)$, then $\mathbf{v}' = (a_1p + u_1, \dots, a_ip - u_i, \dots, a_kp + u_k)$ for some integers $a_1, \dots, a_k$. In this case, pick

$$\mathbf{u}' := (a_1p + u_1, \dots, (l - a_i)p + u_i, \dots, a_kp + u_k) \in \pi^{-1}(\mathbf{u}).$$

If $\mathbf{v} = a\mathbf{u}$ for some $a \in \mathbb{Z}_p^\times$, then by Chinese Remainder Theorem there is $b \in \mathbb{Z}_{lp}$ such that $b \equiv a^{-1} \pmod p$ and $b \equiv 1 \pmod l$. In particular, $b \in \mathbb{Z}_{lp}^\times$ and

$$\mathbf{u}' := b\mathbf{v}' \in \pi^{-1}(\mathbf{u})$$

because $\pi(b\mathbf{v}') = \pi(b)\pi(\mathbf{v}') = a^{-1}(a\mathbf{u}) = \mathbf{u}$.

Then, whichever way $\mathbf{u}'$ satisfies properness, it can be checked that $\mathbf{v}'$ also satisfies properness in the same way (we need b to be a unit because if a time t is a witness for $\mathbf{u}$, then the time $b^{-1}t$ is a witness for $\mathbf{v}$). Therefore every speed tuple in $\pi(\mathbf{v})$ is proper, and thus $\mathbf{v}$ is eventually proper. □

Therefore, we may only verify the eventual properness for a single element in an equivalence class. We implement this by fixing by ordering the speeds ascendingly, limiting the speeds to be in the first half of residues modulo p, and fixing v_1 to be 1. In other words, we check all tuples of integers $(v_1, \dots, v_k)$ with

$$1 = v_1 \le v_2 \le \cdots \le v_k \le \frac{p-1}{2}.$$

This implementation results in k speed tuples per equivalence class, so there is a possibility to improve the run time by a factor of k. However, we are unable to find a way to generate a single element from each equivalence class

We should stress that this reduction is essential to obtaining our results. There are approximately p^k tuples in the $(k, p, 1)$-ansatz, but with this reduction we only need to check a much smaller number of approximately

$$\binom{p/2}{k-1} \approx \frac{p^k}{2^k(k-1)!}.$$

Compared to the implementation of [11] and [14], we reduce the number of tuples to check by a factor of approximately p/k.

5.2. **Verification algorithms.** Fix $k \in \{10, 11, 12\}$ and a prime p. Our goal is to verify that $J(k, p) = \emptyset$ in order to apply Corollary 2.5. To do this, we use a verification algorithm that consists of two phases: computing $I(k, p, 1)$ and repeated lifting.

The first phase is done by choosing representative speed tuples as described in the previous subsection. We use lifting diagram to explain the second phase. In a lifting diagram, the first set S_1 is always the precomputed $I(k, p, 1)$. Two kinds of arrows correspond to two actions from Section 3:

- $S_n \xrightarrow{\times c} S_{n+1}$ indicates lifting from S_n to S_{n+1} with constant c.
- $S_n \xrightarrow{\div d} S_{n+1}$ indicates projecting with $\pi_{dp\to p}$ (this is as same as π_p).

Since we only perform these two actions, Proposition 3.1 inductively asserts that $J(k, p) \subseteq \pi_p S_n$ for every n.

For $k = 11$, since $k + 1 = 12$ only has small prime factors 2 and 3, it is convenient to lift through its prime factors. Hence, the lifting diagram is

$$S_1 \xrightarrow{\times 2} S_2 \xrightarrow{\times 2} S_3 \xrightarrow{\times 2} S_4 \xrightarrow{\times 2} S_5 \xrightarrow{\times 3} S_6 \xrightarrow{\times 3} S_7.$$

If S_7 is empty, we conclude that $J(k, p) \subseteq \pi_p S_7$ is empty.

For $k \in \{10, 12\}$, $k+1$ is prime. In order to avoid the lift with $c = k+1$, we lower the goal of the lifting phase from showing that the final set is empty to showing that it only consists of equivalents of $(1, 2, \ldots, k)$. Therefore, the lifting diagram for both cases is

$$S_1 \xrightarrow{\times 2} S_2 \xrightarrow{\times 2} S_3 \xrightarrow{\times 2} S_4 \xrightarrow{\div 8} S_5 \xrightarrow{\times 3} S_6 \xrightarrow{\times 3} S_7 \xrightarrow{\div 9} S_8$$

which in particular does not contain a $c = 11$ or $c = 13$ lift.

As $k+1$ is prime, Proposition 4.4 asserts that $(1, 2, \ldots, k) \notin J(k, p)$ if $p > k^2 + k$. This is why we only use $p > k^2 + k$ for $k \in \{10, 12\}$.[7] Therefore, if there are only speed tuples equivalent to $(1, 2, \ldots, k)$ remaining in S_8, we can verify the emptiness of $J(k, p)$.

5.3. **Further particulars.** We implement the algorithms in Subsection 5.2 in C++. Compared to the original implementations of [10] and [15], we improve practical performance by refactoring to support parallel execution and by using more memory-efficient data structures. The development of our code was assisted by AI-based tools.

In computing S_1, we modified the depth-first-search technique developed by [10] to account for the equivalent relation mentioned in Subsection 5.1.

These refinements dramatically improve runtime. We verified the case of $k = 8$ in 4 seconds (15 minutes using [15]), $k = 9$ in 90 seconds (23 hours using [15]), and $k = 10$ in 45 minutes on a 10-core Apple M4 processor (excluding compilation time). The proofs of $k = 11$ and $k = 12$ were run in multiple batches, with multiple versions, and on multiple machines, making the exact time taken incomparable to other cases.

The source of the computation, along with the log files, can be found in the first author's GitHub repository [13].

6. Proof of Theorem 1.3

Computer verification shows that, for each $k \in \{10, 11, 12\}$, the set P_k in Table 1 satisfies $J(k, p) = \emptyset$ for every $p \in P_k$. Moreover, for each k, we have that $\prod_{p \in P_k} p \geq B_k$. Since $LRC(9)$ holds [14], we may apply Proposition 2.7 thrice and conclude that $LRC(k)$ holds for $k \in \{10, 11, 12\}$.

7. Concluding remarks

In Section 4, we mentioned that, surprisingly, we only need to perform $c = 2$ and $c = 3$ lifts (each more than once) to eliminate all speed tuples apart from those whose projection modulo p is $(1, 2, \ldots, k)$, which is a *tight* speed tuple. From Goddyn and Wong [7], a speed tuple is said to be tight if the strict inequality cannot hold in the statement of Conjecture 1.2.

An easy continuity argument shows that if $\mathbf{v}$ is not tight, then there is an interval $(T, T + \epsilon)$ such that every $t \in (T, T + \epsilon)$ is a witness time for $\mathbf{v}$. Therefore, for any non-tight speed tuple $\mathbf{v} \in \mathbb{Z}^k_{p,l}$, there is L such that for any $l \geq L$ there is a witness time in $\frac{1}{lp}\mathbb{Z}$. Our observation suggests that there is a universal l for every non-tight $\mathbf{v}$:

Conjecture 7.1. *Fix p prime and $k+1$. There exists a constant L such that, for any integer $l \geq L$, every non-tight speed tuple $\mathbf{v} \in \mathbb{Z}^k_{p,l}$ has a witness time in $\frac{1}{lp}\mathbb{Z}$.*

This might even be a universal bound for the denominator irrespective of the ansatz.

Conjecture 7.2. *Let $k+1$ be a positive integer. There exists a constant D such that, for any integer $d \geq D$, every non-tight speed tuple $\mathbf{v} \in \mathbb{Z}^k_{>0}$ has a witness time in $\frac{1}{d}\mathbb{Z}$.*

On a different note, the algebraic method in Proposition 4.1 may be extensible to verify Conjecture 1.2 for other classes of speed tuples.

Looking ahead, the primary bottleneck in extending our results to $k = 13$ is the efficient computation of $I(k, p, 1)$. Progress here likely requires a better understanding of speed tuples

[7] In fact, we can use some primes $p < k^2 + k$ if $\mathbf{r}_k(\frac{1}{k+1}\mathbb{Z}) \subseteq \mathbf{r}_k(\frac{1}{p}\mathbb{Z})$ by imitating the proof of Lemma 4.3. This inclusion holds for $p \in \{103, 107, 109\}$ when $k = 10$, and $p \in \{149, 156\}$ when $k = 12$, and we have checked that for these k, p, $J(k, p) = \emptyset$.

k	P_k	$\ln \prod_{p \in P_k} p$	$\ln B_k$ (Lemma 2.6)
10	127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467	> 342	< 338
11	23, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499, 503, 509, 521, 523, 541, 547, 557, 563, 569, 571, 577	> 435	< 435
12	167, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499, 503, 509, 521, 523, 541, 547, 557, 563, 569, 571, 577, 587, 593, 599, 601, 607, 613, 617, 619, 631, 641, 643, 647, 653, 659, 661, 673, 677, 683, 691, 701, 709, 719, 727, 733	> 547	< 546

TABLE 1. Sets of primes used in the verification of $LRC(k)$ for $k \in \{10, 11, 12\}$.

that do not have a witness time in an ansatz, which in turn could yield stronger pruning conditions and make the initial sieve more tractable.

ACKNOWLEDGEMENTS

The authors are most thankful to Noah Kravitz for his insightful feedback and suggestions, especially on the exposition of the paper. The second author acknowledges the support from the King's Scholarship (Thailand).

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